SRS: IT Institute Inquiry-to-Enrollment Analytics Project

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1. Introduction

1.1 Purpose & Educational Scope

This SRS defines a capstone project in which students turn 2,500 raw inquiry records from an IT training institute into business insight: where leads come from, how they convert to enrolled students, and how much revenue they bring. Students work through the full analytics chain, from Excel profiling to Python cleaning, MySQL storage, machine learning and a Power BI dashboard.

The project reuses the Revenue Operations (RevOps) approach from B2B sales analytics. An inquiry plays the role of a lead, a counselor plays the sales rep, and an enrollment is a closed-won deal:

RevOps term	Institute equivalent	Source column
Lead	Inquiry	Inquiry_ID
Sales Rep	Counselor	Counselor
Lead Source	Inquiry source	Inquiry_Source
Product / Industry	Course interested	Course_Interested
Deal stage	Status (Follow-up, Demo Attended, Enrolled, Not Interested)	Status
Closed-won deal size	Fee paid (INR)	Fee_Paid_INR
Sales cycle	Enrollment_Date minus Inquiry_Date	Inquiry_Date, Enrollment_Date
Region	City	City

In scope: data cleaning, storage, analysis, prediction and dashboarding on the supplied dataset. Out of scope: a live CRM, real-time data feeds and any use of real student personal data (the dataset is synthetic).

1.2 Core Learning Objectives & Tech Stack

By the end of the project, a student can:

- Profile a dataset and document its quality problems in Excel.
- Clean data, parse dates and encode categories with Python (pandas).
- Design a relational schema and write DDL and analytical queries in MySQL.
- Engineer features and run exploratory analysis with funnel, heatmap and box-plot charts.
- Build a lead-scoring classifier and a revenue-forecasting regressor with scikit-learn.
- Model data, write DAX measures and publish an interactive Power BI dashboard.

Layer	Tool	Used for
Profiling	Microsoft Excel	Quality checks, validation rules
Wrangling	Python 3.10+, pandas, NumPy	Cleaning, date arithmetic, encoding
Analysis and ML	Matplotlib, Seaborn, scikit-learn, Jupyter	EDA, classification, regression
Storage	MySQL 8	Schema, queries, feature tables
BI	Power BI Desktop	Data model, DAX, dashboard
Version control	Git / GitHub	Submission repository

2. The Data Pipeline & Tool Application

The pipeline runs in four steps: Excel profiling, then Python cleaning, then MySQL storage, then analysis. Each phase hands a defined file to the next.

2.1 Data Sources & Extraction

Dataset sourcing. Students receive `it_institute_inquiries_enrollments.csv` (2,500 rows, 12 columns) and keep an untouched copy in `/data/raw`. Source systems modeled: website forms, walk-in registers, social media, ad platforms and campus drives.

Data types to handle:

Column	Type	Notes
Inquiry_ID	String (key)	Format INQ00001; must be unique

Column	Type	Notes
Inquiry_Date	Date	ISO format YYYY-MM-DD
Student_Name	String	Free text; trim whitespace, fix casing
Gender	Categorical	Male / Female
Phone	Integer / String	10 digits, starts with 6 to 9; store as text
City	Categorical	10 cities
Course_Interested	Categorical	9 courses
Inquiry_Source	Categorical	8 channels
Counselor	Categorical	5 counselors
Status	Categorical (target)	Enrolled, Follow-up, Demo Attended, Not Interested
Enrollment_Date	Date (nullable)	Filled only when Status is Enrolled
Fee_Paid_INR	Integer (nullable)	Filled only when Status is Enrolled

2.2 Phase 1: Initial Data Profiling (Excel)

Quality checks

- Count blanks per column and confirm Enrollment_Date and Fee_Paid_INR are blank exactly when Status is not Enrolled.
- Find duplicate Inquiry_IDs and duplicate Phone numbers (COUNTIF).
- Spot dates stored as text, inconsistent casing and stray spaces.
- Flag Enrollment_Date earlier than Inquiry_Date, and any Inquiry_Date in the future.
- Check Fee_Paid_INR for zero, negative or extreme values with a quick pivot and min/max.

Data validation

- Dropdown lists (Data Validation) for Gender, City, Course_Interested, Inquiry_Source, Counselor and Status.
- Date-range rule on the two date columns; whole-number rule on Fee_Paid_INR.
- Phone length rule: exactly 10 digits.
- Conditional formatting to highlight violations.

Deliverable: `data_profiling_report.xlsx` with a Quality Checks sheet (check, rule, rows failed, action), a Validation sheet and a short pivot summary of inquiries by status and course.

2.3 Phase 2: Data Wrangling & Automation (Python)

All steps run in `1_python_data_cleaning_and_date_parsing.ipynb` and write the result to `/data/processed`.

Sales cycle arithmetic. Parse both date columns with `pd.to_datetime(errors="coerce")`, then compute:

```
df["Sales_Cycle_Days"] = (df["Enrollment_Date"] - df["Inquiry_Date"]).dt.days
```

Also derive Inquiry_Month, Inquiry_Quarter and Inquiry_Weekday. Reject negative cycle lengths.

Handling missing values. Blank Enrollment_Date and Fee_Paid_INR for non-enrolled inquiries are structural, not errors: keep them as NaT/NaN and add an `Is_Enrolled` flag (1/0). Impute or drop only genuine gaps (for example a missing City), and log every decision in the notebook.

Categorical encoding. Standardize text, then use one-hot encoding for City, Course_Interested and Inquiry_Source, label encoding for Gender and the Status target, and a frequency or target encoding option for Counselor.

Automation. Wrap the steps in a reusable `clean_data(path)` function so that a new CSV can be cleaned with one call.

2.4 Phase 3: Relational Storage (MySQL)

Schema design. Normalize the flat file into dimension tables and two fact tables:

Table	Key columns	Purpose
dim_course	course_id, course_name, list_fee	Course catalogue
dim_counselor	counselor_id, counselor_name, monthly_target	Counselor roster and quota
dim_source	source_id, source_name	Inquiry channels
dim_city	city_id, city_name	Region
fact_inquiry	inquiry_id, inquiry_date, student_name, gender, phone, city_id, course_id, source_id, counselor_id, status	One row per inquiry

Table	Key columns	Purpose
fact_enrollment	enrollment_id, inquiry_id (FK), enrollment_date, fee_paid_inr	One row per enrolled student

SQL DDL/DML. `CREATE TABLE` with primary keys, foreign keys, `NOT NULL` and `CHECK` constraints; load the cleaned CSV with `LOAD DATA INFILE` or `INSERT ... SELECT` from a staging table; add indexes on date and foreign-key columns.

Advanced querying. Students write and save queries that use:

- JOINs across fact and dimension tables.
- Conversion rate by course, source and counselor (`COUNT`, `CASE WHEN`, `GROUP BY`).
- Window functions: `RANK()` of counselors by revenue, running monthly revenue total, `LAG()` for month-over-month change.
- CTEs for the funnel (inquiry, demo, enrolled) and average sales cycle per course.
- A view `vw_ml_features` that feeds Section 3.

3. Advanced Analytics & Modeling

3.1 Feature Engineering (Python & SQL)

Features are built in `2_eda_and_revops_feature_engineering.ipynb` and mirrored in the SQL view `vw_ml_features`.

Feature	Definition	Built in
Sales_Cycle_Days	Enrollment_Date minus Inquiry_Date (enrolled only)	Python / SQL
Inquiry_Month, Quarter, Weekday	Date parts of Inquiry_Date	Python / SQL
Source_Conversion_Rate	Enrolled inquiries divided by all inquiries, per source	SQL
Counselor_Win_Rate	Enrolled inquiries divided by all inquiries, per counselor	SQL
Course_Demand_Rank	Rank of the course by inquiry count in the month	SQL (window)
Fee_Ratio	Fee_Paid_INR divided by course list fee	Python

Feature	Definition	Built in
Is_Enrolled	1 if Status is Enrolled, else 0 (classification target)	Python

Avoid leakage: Enrollment_Date, Fee_Paid_INR, Sales_Cycle_Days and Fee_Ratio exist only after enrollment, so they are excluded from the lead-scoring features.

3.2 Exploratory Data Analysis (Python)

Each chart needs a title, labeled axes and a one-sentence written insight.

- **Funnel drop-off charts.** Inquiries, then Follow-up, then Demo Attended, then Enrolled, with drop-off percentages; repeat by course and by source.
- **Correlation heatmaps.** Seaborn heatmap of numeric and encoded features against Is_Enrolled and Fee_Paid_INR.
- **Deal size box plots.** Fee_Paid_INR by course and by counselor to show spread, medians and outliers.

3.3 Introductory Machine Learning (Python)

Predictive lead scoring (classification). Notebook `3_lead_scoring_classification_model.ipynb` predicts whether an inquiry will enroll.

- Models: Logistic Regression as the baseline, Random Forest as the comparison.
- Split: 80/20 train-test, stratified on Is_Enrolled, with 5-fold cross-validation.
- Metrics: accuracy, precision, recall, F1, ROC-AUC and a confusion matrix.
- Output: a lead score (probability of enrolling, 0 to 100) per inquiry and the top feature importances.

Revenue forecasting (regression). Aggregate Fee_Paid_INR by month, then forecast the next 3 months.

- Models: Linear Regression with month and seasonality features; optional Random Forest Regressor.
- Split by time (train on earlier months, test on the latest), not at random.
- Metrics: MAE, RMSE and R-squared; plot actual against predicted revenue.

The data is synthetic, so students must state clearly in the notebook that model results show the method, not real market behavior.

4. Business Intelligence & Storytelling (Power BI)

4.1 Dashboard Development

Data modeling in Power BI. Import the MySQL tables and build a star schema: `fact_inquiry` and `fact_enrollment` in the center, joined one-to-many to dim_course, dim_counselor, dim_source, dim_city and a dim_date calendar table. Mark dim_date as the date table and use single-direction relationships.

DAX (Data Analysis Expressions). Minimum measures:

```
Total Inquiries = COUNTROWS(fact_inquiry)
Enrollments = CALCULATE([Total Inquiries], fact_inquiry[status] = "Enrolled")
Conversion Rate = DIVIDE([Enrollments], [Total Inquiries])
YTD Revenue = TOTALYTD(SUM(fact_enrollment[fee_paid_inr]), dim_date[date])
Weighted Pipeline = SUMX(fact_inquiry, fact_inquiry[stage_probability] *
RELATED(dim_course[list_fee]))
Avg Sales Cycle = AVERAGEX(fact_enrollment,
DATEDIFF(RELATED(fact_inquiry[inquiry_date]),
fact_enrollment[enrollment_date], DAY))
Quota Attainment = DIVIDE(SUM(fact_enrollment[fee_paid_inr]),
SUM(dim_counselor[monthly_target]))
```

`stage_probability` is a stage lookup (for example Follow-up 0.2, Demo Attended 0.5) that students define and document.

Visual interface.

- **RevOps KPI ribbon:** executive cards showing Year-to-Date (YTD) revenue, total weighted pipeline and current average sales cycle length.
- **Interactive sales funnel:** a funnel visual showing the conversion rate between each stage (Inquiry, Follow-up, Demo Attended, Enrolled), recalculating on the selected city or counselor.
- **Counselor leaderboard matrix:** a conditionally formatted table of counselor against quota attainment, win rate and total generated pipeline, highlighting top performers.
- **Interactive slicers:** filters for Counselor, City, Course, Inquiry Source and Quarter.

Design requirements: one landing page plus one drill-down page, consistent colors, readable titles, tooltips on charts and a short "key insights" text box on each page.

5. Student Submission & Evaluation Rubric

5.1 Repository Structure

Students submit a logically organized GitHub repository or ZIP file that demonstrates best practices:

```
project-root/
├── README.md
├── data/
|   ├── raw/          original files, never edited
|   └── processed/    cleaned files ready for MySQL
├── scripts/
|   ├── 1_python_data_cleaning_and_date_parsing.ipynb
|   ├── 2_eda_and_revops_feature_engineering.ipynb
|   └── 3_lead_scoring_classification_model.ipynb
├── sql/              DDL (table creation) and analysis queries
└── dashboard/        finalized .pbix Power BI file
```

The README states the project goal, how to run each notebook, the order of execution and the main findings.

5.2 Evaluation Criteria

Criterion	Weight	What is assessed
Code Quality & Documentation	30%	Readable code, comments and markdown explanations, reproducible notebooks, clear README, sensible naming and structure
Data Engineering	30%	Correct cleaning and date parsing, missing-value handling, sound schema design, working DDL and queries, clean load into MySQL
Analytical Accuracy	20%	Correct metrics and DAX measures, valid features without leakage, properly evaluated models, insights supported by the data
Dashboard & Storytelling	20%	Clear KPI ribbon, working funnel, leaderboard and slicers, design quality, and a concise business narrative

The first three weights follow the brief; the 20% for Dashboard & Storytelling is added so that the total reaches 100%. Adjust it if your course assigns the remaining 20% differently.